IS353IA · ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING · LAB PART B

Lateralis

User manual and technical overview

Temporal Graph Learning for Cyber Attack-Path Forecasting and Adaptive Defense

Student	Krrish Agarwal
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Repository	github.com/krish-31/aiml-sem5
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In one sentence

Lateralis watches a company network as a graph that changes every hour, forecasts the attacker's next hop after a break-in, ranks the likely routes to critical servers and recommends the cut that blocks them.

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1. What Lateralis does

Serious attacks are rarely a single event. An attacker lands on one workstation, steals the passwords of people who log in there, and hops machine to machine (lateral movement) until they reach a domain controller or database. Most detection tools judge each login on its own and raise the alarm late. Lateralis instead:

- **Observes** login logs as one graph per hour: users and machines are nodes, logins are edges.
- **Forecasts** the attacker's next hop with a temporal graph neural network, as calibrated probabilities.
- **Ranks** the most likely complete routes from the compromised machines to critical assets.
- **Recommends** a response: routine containment of the known compromised machine, plus the next cut, the access removal that best blocks the route ahead for the least disruption.
- **Explains** itself: a live event feed, a full event log with flags, and a one-click PDF incident report.

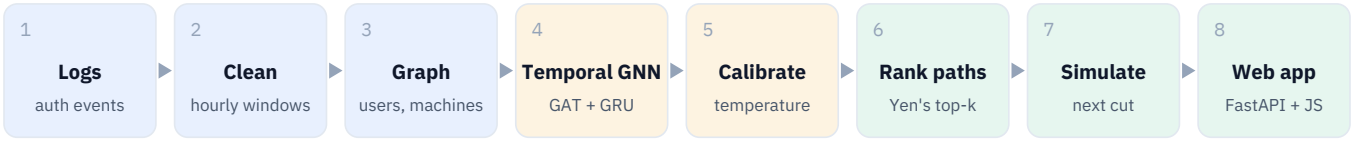


Figure 1. The eight-stage pipeline, from raw logs to the web app.

All results in this document come from 300 synthetic company networks (Section 8). Real datasets are the next milestone (Section 11).

2. Quick start: install and run

Requirements: macOS, Linux or Windows with Python 3.11 and uv; about 2 GB of disk. No GPU is needed for the demo. Training takes about 15–25 minutes on a laptop CPU.

2.1 One-time setup

```
git clone https://github.com/krish-31/aiml-sem5 && cd aiml-sem5
uv venv --python 3.11 .venv
uv pip install --python .venv/bin/python -e ".[dev,ml,app]"
```

2.2 Build data and model (skip if data/ and checkpoints/ already exist)

```
.venv/bin/python -m attackpath.data.synthetic --n-graphs 300   # 10 s
.venv/bin/python -m attackpath.graph.builder                   # 15 s
.venv/bin/python -m attackpath.models.train                    # 15-25 min
.venv/bin/python -m attackpath.eval.evaluate                   # 2 min
```

2.3 Start the app

```
.venv/bin/python -m attackpath.api.server
```

Open **http://127.0.0.1:8000** in a browser. The first visit to the console takes a few seconds while the model replays the demo network; after that every step is instant. Everything runs locally: no internet is needed, fonts are bundled.

> **Before presenting**
>
> Start the server 5 minutes early, open the Live console once to warm it up, and use full-screen (F11 or Ctrl+Cmd+F). A 1920×1080 or 1280×720 screen both work.

3. How to use the app

The menu has four pages: **01 Overview**, **02 Live console**, **03 Results** and **04 Method**. The button at the top right switches between dark and light themes and remembers your choice. Two further views open from inside the console: the full **Event log** and **Reports**.

3.1 Overview

The landing page. Its right-hand panel is not an animation: it replays a real attack on NET-062, a network the model never saw in training, and prints the model's live output as an intercept log (ALERT, FCAST = forecast, PATH, ACTION, HOP = what the attacker actually did and the rank the model had given it). Below it are the four headline results and the research-gap table comparing Lateralis with prior work.

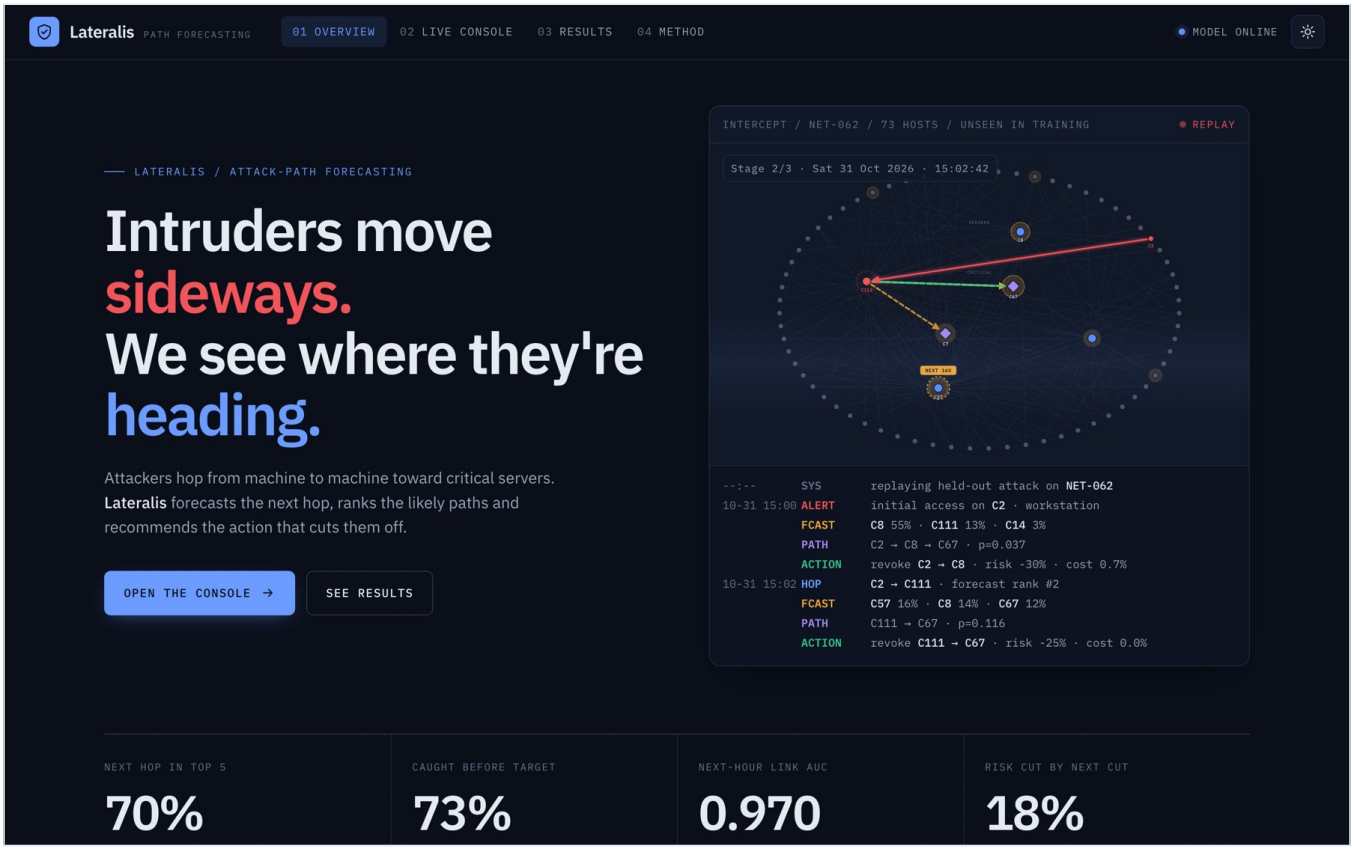


Figure 2. Overview page with the live intercept replay.

3.2 Live console

The main demo screen. It replays an attack on a held-out network one hop at a time and shows what the model forecasts at each moment.

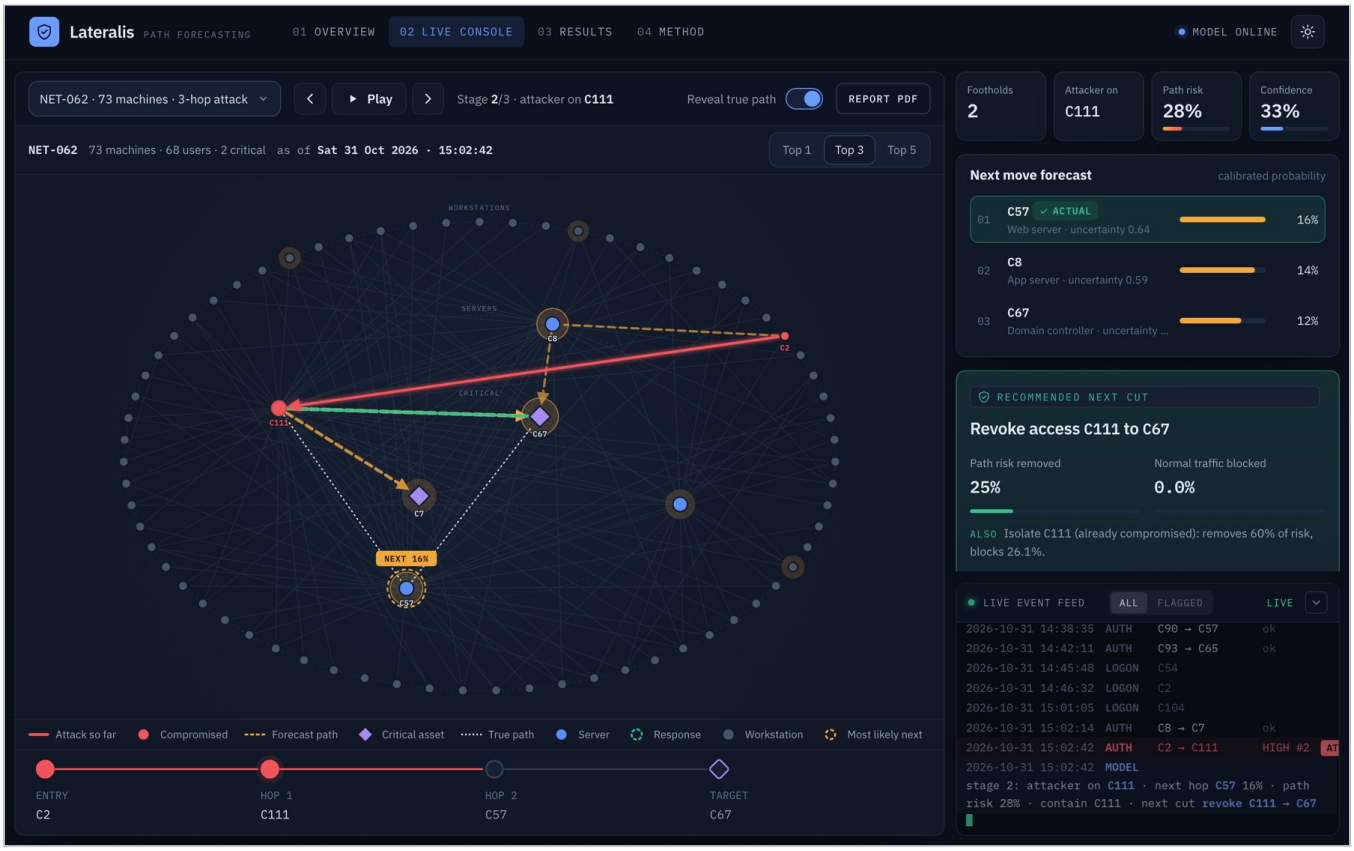


Figure 3. Live console at stage 2 of NET-062 with the true path revealed. The attacker moved C2 → C111; the model forecasts C57 next (16%, marked NEXT) and recommends revoking C111 → C67.

Toolbar (top)

Control	What it does
Network selector	Pick one of the 45 test networks (never seen in training). NET-062 is the default demo.
Back · Play · Next	Step back, play automatically (one hop every 2.6 s) or step forward. The left and right arrow keys also work.
Stage label	Which stage you are on and where the attacker is now.
Reveal true path	Shows where the attacker really went (dotted line, ACTUAL badges, timeline filled in).
Report PDF	Opens the incident report for the current stage in a new tab.
as of ...	The exact moment the screen describes: the last hour of logs read, or the observed hop.
Top 1 / 3 / 5	How many forecast paths to draw on the map.

Map (centre)

Critical assets (purple diamonds) sit in the middle ring, servers (blue) in the second ring, workstations (grey) outside. Red lines are the attack so far, amber dashed lines the forecast paths, the green dashed line the recommended cut, and the amber **NEXT p%** tag marks the single most likely next machine. Hover any machine to see its next-hop probability. The legend sits in its own strip under the map so it never covers the network.

Timeline (bottom)

Entry → hops → target. Red steps have happened. Amber steps are forecasts: the most likely next machine with its probability, and the likely target. Click any step to jump to it.

Side panel (right)

Panel	Meaning
Footholds	How many machines the attacker controls.
Attacker on	The machine the attacker is on now.
Path risk	Combined probability of the top routes to a critical asset.
Confidence	How focused the next-move forecast is (100% = one clear answer).
Next move forecast	Top 3 machines the attacker is likely to move to next, with calibrated probabilities. Uncertainty is the entropy of that single yes/no guess (0 = sure, 1 = coin flip).
Recommended next cut	The action that removes the most risk on the attacker's routes from where they are now, for the least normal traffic blocked. The line below it shows routine containment (isolate the known compromised machine) with its own numbers.
Live event feed	A tail of the authentication log with real timestamps. HIGH #2 = a login from a compromised machine to the 2nd-ranked forecast destination; MED = any other login from a compromised machine or a very unexpected one. MODEL lines show each new forecast. All/Flagged filters, a minimize button with a new-events counter, and a 'full log' link.

Deep links

You can open the console at a given moment, e.g. `#/console?net=62&stage=2&reveal=1` (stage is 1-based).

3.3 Event log (from the feed's 'full log' link)

Every machine-to-machine login around the attack with search, All / Flagged / High filters and a Reveal attack switch. Columns: time, user, connection, signals (from compromised, new route), forecast rank, 'unexpected' score from the model's link head, and the flag.

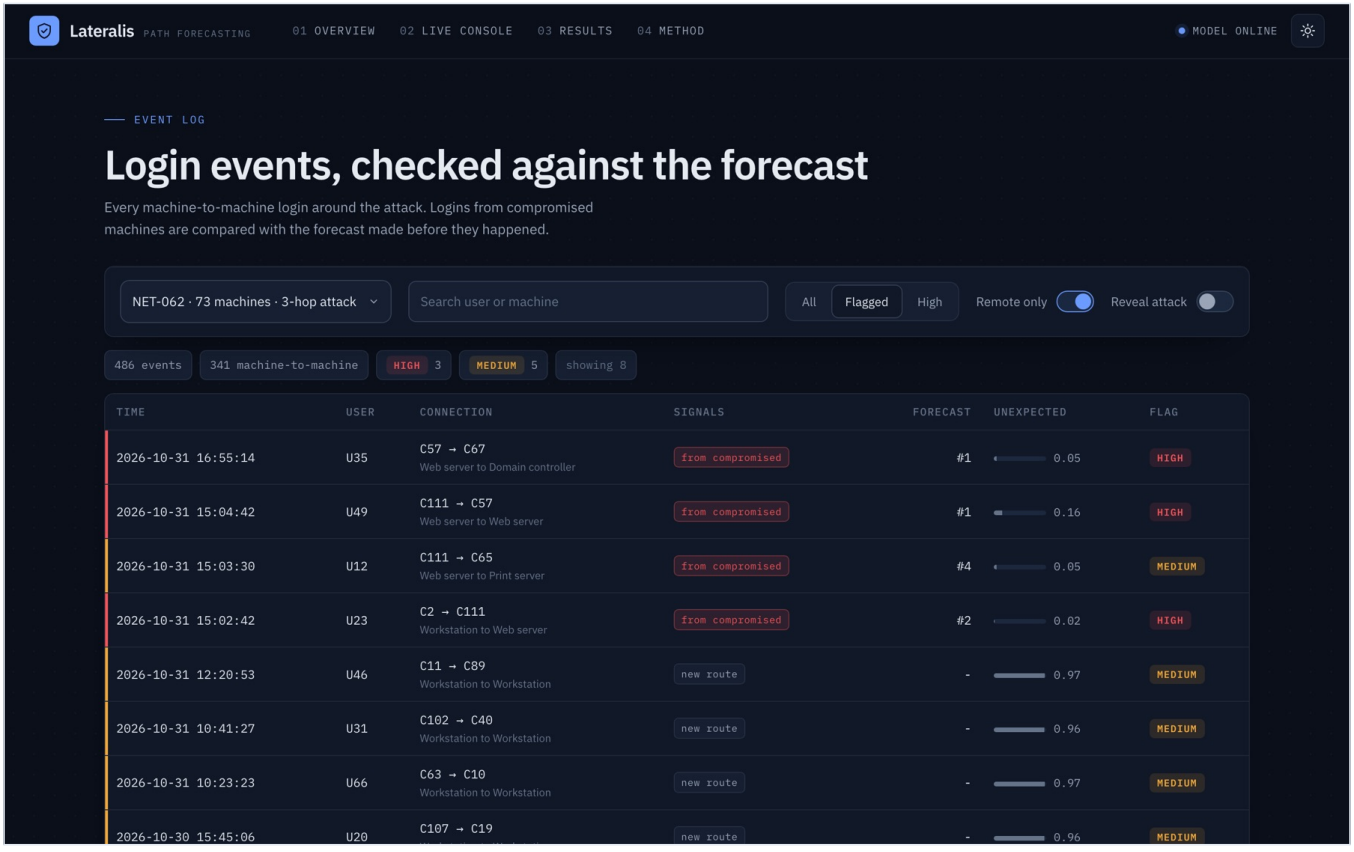
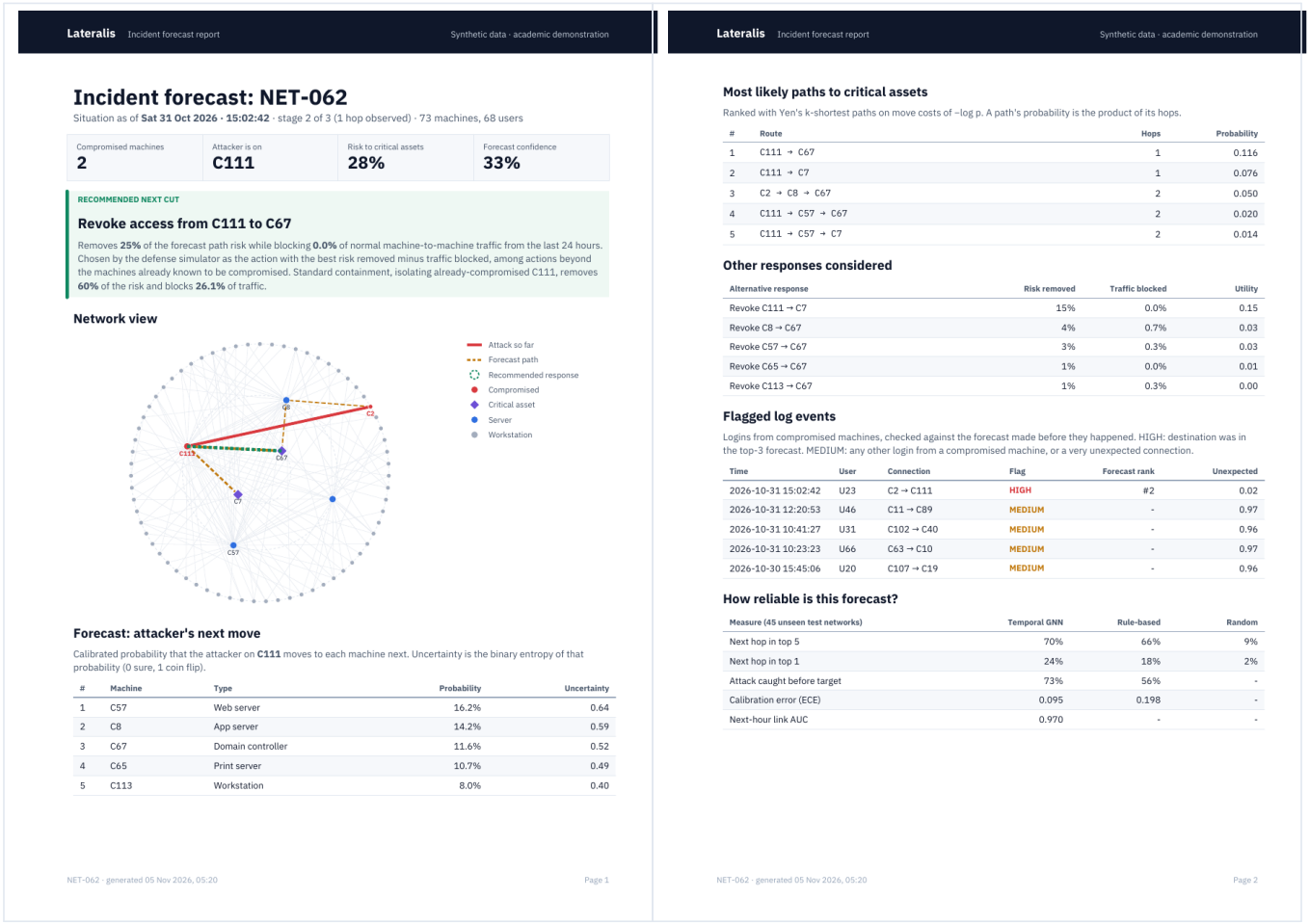


Figure 4. Event log for NET-062: the attacker's three logins are all flagged HIGH.

3.4 Reports and the PDF incident report

The console's **Report PDF** button (or the Reports view at `#/reports`) produces a two-to-three page PDF for any network and stage: summary, recommended next cut and containment, network diagram, next-move forecast, top paths, other responses, flagged events known at that moment (never later ones), model reliability, and an optional ground-truth appendix.



Lateralis Incident forecast report — Synthetic data · academic demonstration

Incident forecast: NET-062

Situation as of **Sat 31 Oct 2026 · 15:02:42** · stage 2 of 3 (1 hop observed) · 73 machines, 68 users

Compromised machines	Attacker is on	Risk to critical assets	Forecast confidence
2	C111	28%	33%

RECOMMENDED NEXT CUT

Revoke access from C111 to C67

Removes **25%** of the forecast path risk while blocking **0.0%** of normal machine-to-machine traffic from the last 24 hours. Chosen by the defense simulator as the action with the best risk removed minus traffic blocked, among actions beyond the machines already known to be compromised. Standard containment, isolating already-compromised C111, removes **60%** of the risk and blocks **26.1%** of traffic.

Network view

Forecast: attacker's next move

Calibrated probability that the attacker on **C111** moves to each machine next. Uncertainty is the binary entropy of that probability (0 sure, 1 coin flip).

#	Machine	Type	Probability	Uncertainty
1	C57	Web server	16.2%	0.64
2	C8	App server	14.2%	0.59
3	C67	Domain controller	11.6%	0.52
4	C65	Print server	10.7%	0.49
5	C113	Workstation	8.0%	0.40

NET-062 · generated 05 Nov 2026, 05:20 — Page 1

Lateralis Incident forecast report — Synthetic data · academic demonstration

Most likely paths to critical assets

Ranked with Yen's k-shortest paths on move costs of −log p. A path's probability is the product of its hops.

#	Route	Hops	Probability
1	C111 → C67	1	0.116
2	C111 → C7	1	0.076
3	C2 → C8 → C67	2	0.050
4	C111 → C57 → C67	2	0.020
5	C111 → C57 → C7	2	0.014

Other responses considered

Alternative response	Risk removed	Traffic blocked	Utility
Revoke C111 → C7	15%	0.0%	0.15
Revoke C8 → C67	4%	0.7%	0.03
Revoke C57 → C67	3%	0.3%	0.03
Revoke C65 → C67	1%	0.0%	0.01
Revoke C113 → C67	1%	0.3%	0.00

Flagged log events

Logins from compromised machines, checked against the forecast made before they happened. HIGH: destination was in the top-3 forecast. MEDIUM: any other login from a compromised machine, or a very unexpected connection.

Time	User	Connection	Flag	Forecast rank	Unexpected
2026-10-31 15:02:42	U23	C2 → C111	HIGH	#2	0.02
2026-10-31 12:20:53	U46	C11 → C89	MEDIUM	-	0.97
2026-10-31 10:41:27	U31	C102 → C40	MEDIUM	-	0.96
2026-10-31 10:23:23	U66	C63 → C10	MEDIUM	-	0.97
2026-10-30 15:45:06	U20	C107 → C19	MEDIUM	-	0.96

How reliable is this forecast?

Measure (45 unseen test networks)	Temporal GNN	Rule-based	Random
Next hop in top 5	70%	66%	9%
Next hop in top 1	24%	18%	2%
Attack caught before target	73%	56%	-
Calibration error (ECE)	0.095	0.198	-
Next-hour link AUC	0.970	-	-

NET-062 · generated 05 Nov 2026, 05:20 — Page 2

Figure 5. Pages 1 and 2 of a generated incident report (NET-062, stage 2).

3.5 Results and Method

Results shows the headline numbers, next-hop accuracy and attack-path charts against a rule-based baseline and random guessing, the full table and a reference card. **Method** summarises the architecture, data, training, path ranking and defense simulation.

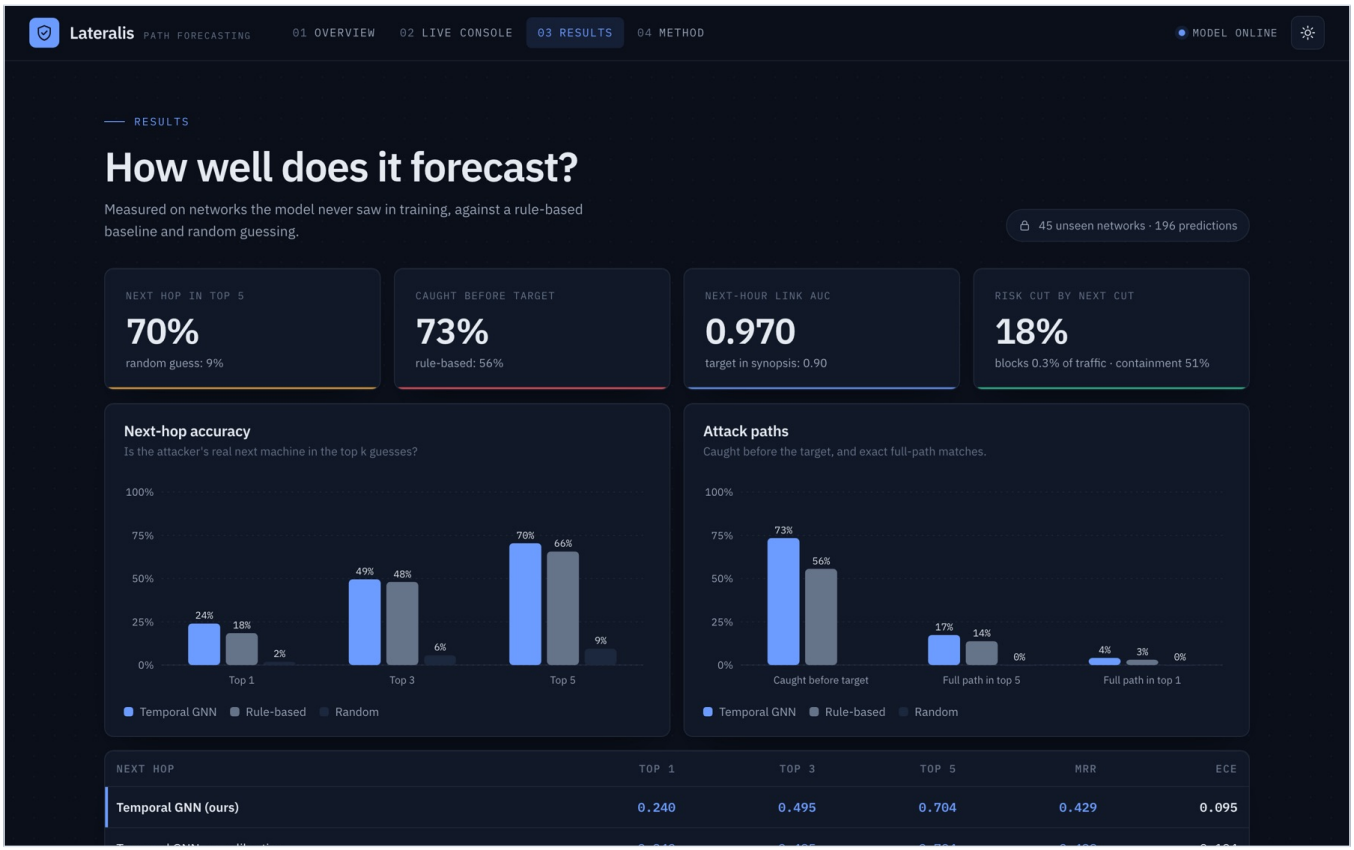


Figure 6. Results page.

4. 5-minute presentation script

Time	Where	What to say / do
0:00	Overview	"Attackers move sideways. India's average breach takes 263 days to contain. Tools judge logins one by one; I forecast the next hop instead." Point at the live replay.
0:45	Overview	Point at the research-gap table: prior work covers one or two of the five abilities; Lateralis covers all five.
1:15	Console, stage 1	"C2 is infected. The model gives C8 55%, marked NEXT. Containment: isolate C2. Next cut: revoke C2 → C8 for 0.7% of traffic."
2:00	Console, Play	Watch the feed: the attacker's login C2 → C111 appears flagged HIGH #2, then the MODEL line updates the forecast.
2:45	Reveal	Turn on Reveal true path: C57 is marked ACTUAL (the #1 guess at stage 2). The timeline fills in.
3:15	Report PDF	Open the report for the stage: everything an incident team needs on two pages.
3:45	Results	70% next hop in top 5 (random 9%), 73% of attacks caught before the target, link AUC 0.970.
4:30	Wrap up	Honest limits (synthetic data, full-path accuracy) and next steps: LANL, baselines, ablations.

5. System architecture

Three layers. The browser renders everything itself from JSON; the FastAPI server holds the trained model in memory and answers requests; the Python package contains all machine learning, simulation and reporting code.

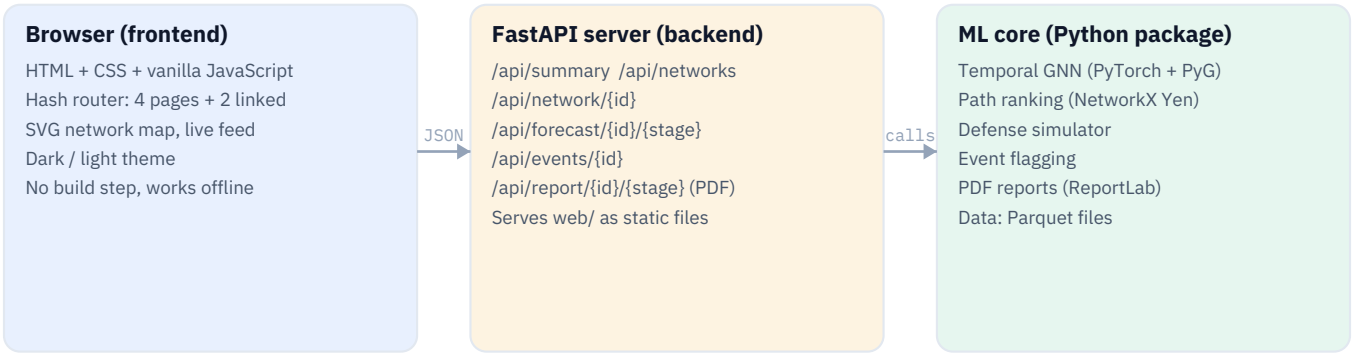


Figure 7. Frontend, backend and ML core.

API endpoints

Endpoint	Returns
`GET /api/summary`	Evaluation results, training history, data split, model size
`GET /api/networks`	The 45 held-out test networks and the default demo network
`GET /api/network/{id}`	Machines with roles and map positions, attack length, exact time of each stage
`GET /api/forecast/{id}/{stage}`	Footholds, next-hop probabilities, top paths, containment, next cuts, recent traffic, ground truth
`GET /api/events/{id}`	Login events around the attack with unexpectedness, forecast rank and flag
`GET /api/report/{id}/{stage}`	PDF incident report (?truth=0 hides the appendix, ?download=1 saves it)

Request flow for one console step

- The browser asks for `/api/forecast/62/1`.
- The server replays NET-062's hourly snapshots through the GNN up to the stage's moment (cached after the first call).
- It turns the hop scores into calibrated move probabilities, ranks paths with Yen's algorithm, and runs the defense simulator over candidate actions.
- The browser draws the SVG map, side panel and timeline, and streams the new events into the feed.

6. Technology stack

6.1 Backend and machine learning

Technology	Version	Used for
Python	3.11.14	Everything server-side
PyTorch	2.15.0	Tensors, training loop, GRU memory, MLP heads
PyTorch Geometric	2.8.1	Heterogeneous graph attention (GATConv, HeteroConv), HeteroData snapshots
NetworkX	3.6.1	Yen's k-shortest simple paths for attack-path ranking
pandas + PyArrow	3.1.1 / 26.0.1	Event tables, Parquet storage, feature aggregation

Technology	Version	Used for
NumPy	2.4.6	Synthetic data generation, metrics
scikit-learn	1.9.1	ROC-AUC and average precision
FastAPI	0.162.0	REST API and static file serving
Uvicorn	0.54.0	ASGI web server
ReportLab	5.1.0	PDF incident reports and this manual
pytest + httpx	9.2.0	25 automated tests (data, graphs, paths, defense, API, timing)

6.2 Frontend

Technology	Used for
HTML5 + CSS3	Layout with CSS grid and flexbox; theme tokens as CSS custom properties (dark and light)
Vanilla JavaScript (ES2020)	Single-page app with a hash router; no framework and no build step
Inline SVG	Network map, charts, icons and the overview replay, all drawn in code
IBM Plex Sans / Mono	Typography, self-hosted (SIL Open Font License), so the app works offline
localStorage	Remembers theme and whether the event feed is minimized

6.3 Tooling

Tool	Used for
uv	Virtual environment and fast dependency installs
Git + GitHub	Version control (github.com/krish-31/aiml-sem5)
Headless Google Chrome	Screenshots in this manual (scripts/screenshots.sh)
Claude (Anthropic)	AI assistant used to help write code and documentation, as disclosed in the synopsis

Change from the synopsis

The synopsis planned a Streamlit front end. It was replaced with a custom HTML/JavaScript front end on FastAPI for a faster, more presentable interface; a Streamlit version is kept in app/dashboard.py as a backup.

7. The model

Part	Design
Input	One heterogeneous graph per hour: computer and user nodes; 'connects' (computer → computer) and 'logs on' (user → computer) edges plus reverse edges. 10 computer features and 8 user features (activity counts, distinct partners, first-time connections, critical flag, time of day, weekend).
Encoder	2 heterogeneous graph-attention layers (GAT), 4 heads, hidden size 64, residual connections, LayerNorm.
Memory	One GRU cell per node type carries each node's state from hour to hour.
Link head	2-layer MLP on two node states plus history features: will u → v appear next hour?
Next-hop head	2-layer MLP scored over every machine, softmax: where does the attacker on u go next?
History features	Seen before, shared users (whose stolen passwords would work), past contact counts, credential value (the most powerful account seen on a machine), in-degree.
Training	Binary cross-entropy with negative sampling (link) + cross-entropy (next hop, weight 2); Adam, lr 0.001, dropout 0.2, gradient clipping, early stopping (patience 6). 145,666 parameters; about 15 minutes on a laptop CPU.
Calibration	Temperature scaling on the validation set ($T \approx 1.01$: the model was already well calibrated).
Path ranking	Move cost $-\log p$; Yen's k-shortest simple paths from the footholds to critical assets; path probability = product of hop probabilities.
Defense simulator	For each candidate action (isolate a machine, revoke one access) re-normalise the move probabilities, re-rank the paths, utility = risk removed − traffic blocked. Containment of known compromised machines is reported separately; the next cut prefers actions on routes from the attacker's current machine.
Event flags	HIGH: login from a compromised machine to a top-3 forecast destination. MEDIUM: other logins from compromised machines, or unexpectedness ≥ 0.95 from the link head.

No information from the future is used: a forecast for hour t + 1 uses only hours up to t, and every screen, feed and report is cut off at the exact moment of its stage (checked by an automated test).

8. The dataset

Public logs (LANL, CICIDS2017, UNSW-NB15) label single attack events but never the complete route an attacker took. To measure path forecasting, Lateralis generates synthetic company networks with labelled attack paths (`src/attackpath/data/synthetic.py`, seed 42).

Property	Value
Networks	300 (train 210 / validation 45 / test 45, split by network)
Machines per network	32–82 (workstations 25–70, servers 4–10, critical assets 2–3)
Time span	7 days per network, hourly snapshots (168 per network)
Login events	1,398,935 in total, of which 2,264 attack events (0.16%)
Attack hops	1,337 hops; 3–6 per attack; targets: 168 databases, 132 domain controllers
Normal behaviour	Office-hour peaks, quieter weekends, hot-desking (6%), admins on jump hosts (30%), rare unusual logins (1%), service accounts linking app servers to databases
Attacker behaviour	Credential theft at each machine, preference for machines where admins log in and for servers (MITRE ATT&CK T1003), retries on the same hop

Files (folder `dataset/`)

File	Contents
`networks.csv`	One row per network: size, split, entry, target, full attack path, start and end time

File	Contents
`nodes.csv`	32,552 machines and users with role and critical-asset flag
`attack_hops.csv`	1,337 attack hops: timestamp, stolen account, from, to
`events_NET-062.csv`	All 5,709 login events of the demo network
`events_all.csv.gz`	All 1.4 million login events (12.5 MB, kept out of Git)
`README.md`	Data dictionary and how the data is generated

Sample: the NET-062 attack

Entry C2, target C67 (domain controller), path `C2 -> C111 -> C57 -> C67`, test split.

Hop	Timestamp	Account	From	To
1	`2026-10-31 15:02:42`	`U23`	`C2`	`C111`
2	`2026-10-31 15:04:42`	`U49`	`C111`	`C57`
3	`2026-10-31 16:55:14`	`U35`	`C57`	`C67`

9. Results

Measured on the 45 held-out test networks (196 next-hop predictions), under identical conditions for every method.

Measure	Temporal GNN	Rule-based	Random
Next hop in top 1	24%	18%	2%
Next hop in top 3	49%	48%	6%
Next hop in top 5	70%	66%	9%
Calibration error (ECE, lower is better)	0.095	0.198	–
Attack caught before target	73%	56%	–
Full remaining path in top 5	17%	14%	0%
Next-hour link forecast AUC	0.970	–	–

Response	Risk removed	Normal traffic blocked	Blocks the real route
Next cut (recommended)	18%	0.3%	20%
Containment (isolate compromised machine)	51%	6.4%	61%

Event flags: every attack login on the test networks is flagged (a third as HIGH), against 3% of other logins.

How to read these numbers

- Next-hop forecasting is about 8× better than random and beats the rule-based baseline on every measure. A linear model on the same features reaches similar top-5 accuracy, so the remaining error mostly comes from randomness in the synthetic attacker.
- Exact full-path accuracy is low because a path needs 3–6 correct guesses in a row; 'caught before target' is the more useful path measure for defenders.
- The next cut is a precise, low-disruption action; containment is the stronger blunt one. Used together: contain, then cut.

10. Limitations

- All results are on synthetic data; behaviour on real enterprise logs is not yet measured.
- Full-path top-5 accuracy (17%) is far below the synopsis target of 70%.
- The replay assumes the initial compromise is detected and that each hop is seen when it happens.
- The defense simulator changes only the model's graph, never real systems; costs are measured on 24 hours of traffic.
- Only one model run (seed 0) so far; the synopsis plans three seeds and walk-forward validation.

11. Next steps

Phase	Step	Detail
Phase 1	Real data: LANL	Stream the 1.6 billion LANL authentication events in chunks, keep the window around the red-team activity, map to the same event format, split by time 70/15/15.
Phase 2	CICIDS2017 and UNSW-NB15	Machine-only flow graphs to test generalisation to a second kind of data.
Phase 3	Baselines from the synopsis	Logistic regression on node embeddings, single-snapshot GAT, Euler-style GCN + GRU.
Phase 4	Ablations and rigour	Remove the GRU and the calibration step one at a time; 3 seeds; 5-fold walk-forward validation; fairness check of false positives for admins vs normal users.
Phase 5	Better path forecasts	Multi-step rollout (re-run the model on predicted hops) and path-level training to lift full-path accuracy toward the target.
Phase 6	Product	Stream live logs into the API, alerting, Docker packaging, role-based access.
Phase 7	Write-up	Final report and a conference paper draft (IEEE / Springer, Scopus-indexed).

A. Project structure and commands

```
src/attackpath/
  data/synthetic.py      synthetic networks and attacks
  graph/builder.py       hourly heterogeneous snapshots
  models/tgnn.py         temporal GNN (GAT + GRU, two heads)
  models/train.py        training + temperature scaling
  paths/rank.py          Yen's k-shortest attack paths
  defense/simulator.py   containment and next-cut search
  eventlog.py            path-aware event flags
  report.py              PDF incident reports
  eval/evaluate.py       test-set metrics and baselines
  api/server.py          FastAPI backend
web/                     frontend (index.html, styles.css, app.js, fonts/)
dataset/                 exported CSV files
docs/                    this manual and its build script
tests/                   25 automated tests

.venv/bin/python -m pytest -q             # run all tests
.venv/bin/python scripts/export_dataset.py # re-export the CSV dataset
sh scripts/screenshots.sh                  # re-capture screenshots (server running)
.venv/bin/python docs/build_manual.py      # rebuild this PDF
```